

# Lecture 15 - Class Activity ANCOVA

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## Lecture 15: Analysis of Covariance (ANCOVA)

### What is ANCOVA?

ANCOVA (Analysis of Covariance) combines regression and ANOVA to: - Compare group means while adjusting for a continuous covariate - Increase statistical power by reducing residual error - Control for confounding variables

### When to Use ANCOVA

Use ANCOVA when you have: - **Response variable:** Continuous - **Predictor variable:** Categorical (factor/groups) - **Covariate:** Continuous variable that affects the response

### Key Assumptions of ANCOVA

1. **Independence** of observations
2. **Normality** of residuals
3. **Homogeneity of variances** across groups
4. **Linearity** between response and covariate within each group
5. **Homogeneity of slopes** (most critical!) - regression slopes must be equal across all groups

#### ! Critical First Step

Always test for **homogeneity of slopes** before proceeding with ANCOVA. If slopes differ significantly between groups, standard ANCOVA is inappropriate.

## Part 1: Cricket Chirping Analysis

### Data Overview

We want to compare chirping rate of two cricket species: - *Oecanthus exclamationis* - *Oecanthus niveus*

But we measured rates at different temperatures, and there's a relationship between pulse rate and temperature. ANCOVA lets us adjust for temperature effect to get a more powerful test!

```
# Create simulated cricket data based on lecture example
set.seed(456)
n <- 40
species <- rep(c("0. exclamationis", "0. niveus"), each = n/2)
temp <- c(rnorm(n/2, mean = 22, sd = 2), rnorm(n/2, mean = 24, sd = 2))
chirp_rate <- 40 + 2.5 * (temp - 23) + ifelse(species == "0. exclamationis", 10, 0) + rnorm(n, sd = 3)
cricket_data <- data.frame(species = species, temp = temp, chirp_rate = chirp_rate)

# View data structure

head(cricket_data)
```

```

      species    temp chirp_rate
1 0. exclamationis 19.31296  40.69557
2 0. exclamationis 23.24355  51.78799
3 0. exclamationis 23.60175  50.75553
4 0. exclamationis 19.22222  40.80589
5 0. exclamationis 20.57129  50.16484
6 0. exclamationis 21.35188  46.24225

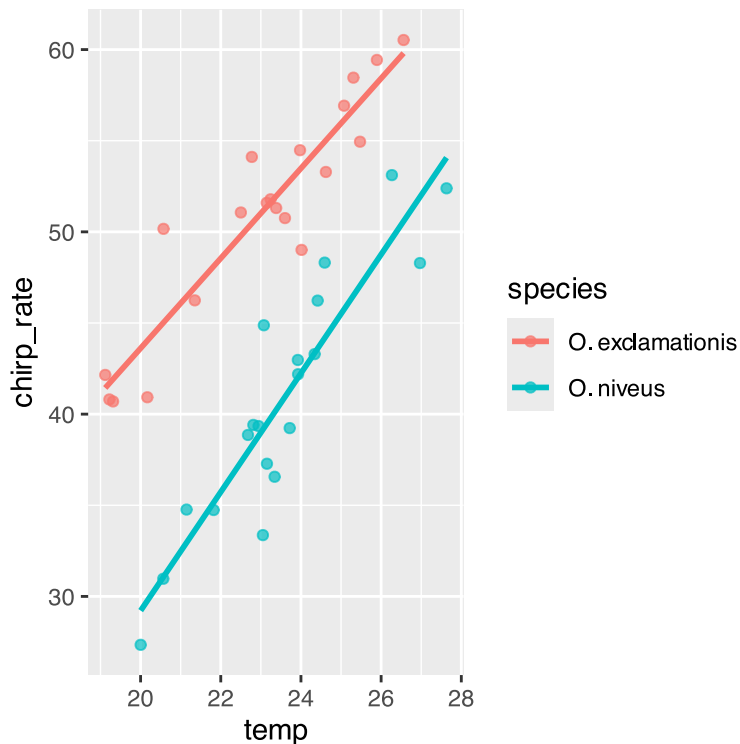
```

```

# Plot with regression lines by species
ggplot(cricket_data, aes(x = temp, y = chirp_rate, color = species)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE)

```

```
`geom_smooth()` using formula = 'y ~ x'
```



## Step 1: Test Homogeneity of Slopes

This is the most critical assumption! We test if the regression slopes are equal across all groups.

```

# Test for homogeneity of slopes by including interaction term
cricket_slopes_model <- lm(chirp_rate ~ temp * species, data = cricket_data)
Anova(cricket_slopes_model, type = 3)

```

Anova Table (Type III tests)

Response: chirp\_rate

|             | Sum Sq | Df | F value | Pr(>F)                |
|-------------|--------|----|---------|-----------------------|
| (Intercept) | 6.32   | 1  | 0.9393  | 0.338915              |
| temp        | 620.48 | 1  | 92.1572 | 0.000000000001828 *** |
| species     | 69.76  | 1  | 10.3617 | 0.002724 **           |

```
temp:species 26.08 1 3.8734 0.056796 .
Residuals 242.38 36
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

**Interpretation:** If  $p > 0.05$ , slopes are homogeneous and we can proceed with ANCOVA. If  $p < 0.05$ , slopes differ and standard ANCOVA is inappropriate.

## Step 2: Fit ANCOVA Model

Since slopes are homogeneous ( $p > 0.05$ ), we can fit the ANCOVA model without the interaction term.

```
# Fit ANCOVA model (without interaction)
cricket_ancova <- lm(chirp_rate ~ temp + species, data = cricket_data)

# Get model summary
summary(cricket_ancova)
```

```
Call:
lm(formula = chirp_rate ~ temp + species, data = cricket_data)

Residuals:
    Min       1Q   Median       3Q      Max
-6.0065 -1.9653  0.1923  0.7886  5.9192

Coefficients:
              Estimate Std. Error t value      Pr(>|t|)
(Intercept)   -13.2012     4.7423  -2.784    0.00842 **
temp             2.7926     0.2048  13.634 0.000000000000000530 ***
species0. niveus -11.8005     0.8593 -13.733 0.000000000000000424 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.694 on 37 degrees of freedom
Multiple R-squared:  0.8994,    Adjusted R-squared:  0.894
F-statistic: 165.4 on 2 and 37 DF,  p-value: < 0.00000000000000022
```

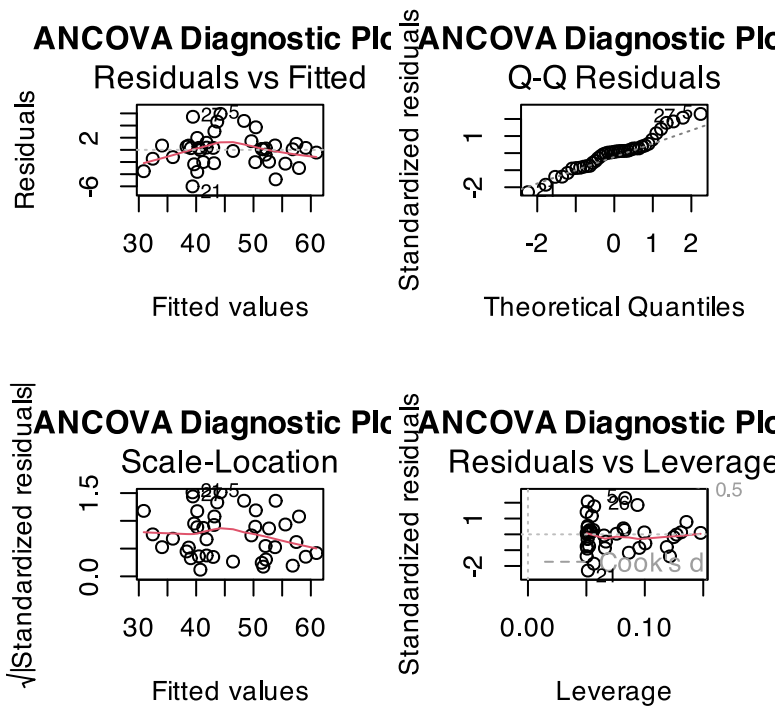
```
# View ANOVA table
Anova(cricket_ancova)
```

```
Anova Table (Type II tests)

Response: chirp_rate
              Sum Sq Df F value    Pr(>F)
temp         1348.81  1  185.90 0.0000000000000005296 ***
species      1368.34  1  188.59 0.0000000000000004236 ***
Residuals    268.46 37
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

## Step 3: Check Model Assumptions

```
# Create diagnostic plots
par(mfrow = c(2, 2))
plot(cricket_ancova, main = "ANCOVA Diagnostic Plots")
```



```
par(mfrow = c(1, 1))
```

## Step 4: Calculate Adjusted Means

ANCOVA compares adjusted means - what each group's mean would be at the overall mean of the covariate.

```
# Calculate adjusted means using emmeans
cricket_adjusted_means <- emmeans(cricket_ancova, "species")

# Convert to dataframe for plotting
cricket_adj_means_df <- as.data.frame(cricket_adjusted_means)
cricket_adj_means_df
```

| species          | emmean   | SE        | df | lower.CL | upper.CL |
|------------------|----------|-----------|----|----------|----------|
| 0. exclamationis | 51.70513 | 0.6049702 | 37 | 50.47934 | 52.93091 |
| 0. niveus        | 39.90462 | 0.6049702 | 37 | 38.67883 | 41.13040 |

Confidence level used: 0.95

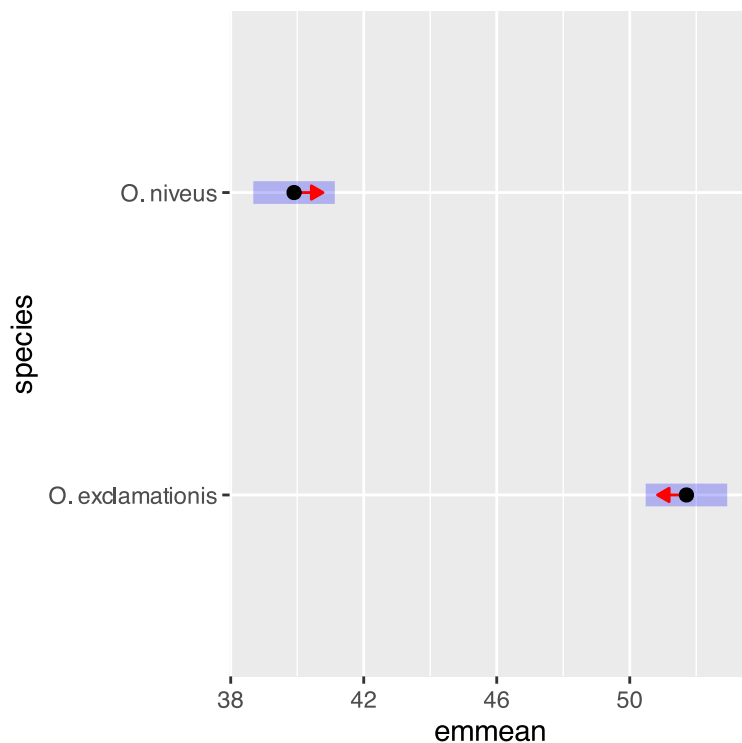
## Step 5: Pairwise Comparisons

```
# Pairwise comparisons of adjusted means
pairs(cricket_adjusted_means, adjust = "sidak")
```

| contrast                     | estimate | SE    | df | t.ratio | p.value |
|------------------------------|----------|-------|----|---------|---------|
| 0. exclamationis - 0. niveus | 11.8     | 0.859 | 37 | 13.733  | <.0001  |

## Step 6: Visualize Results

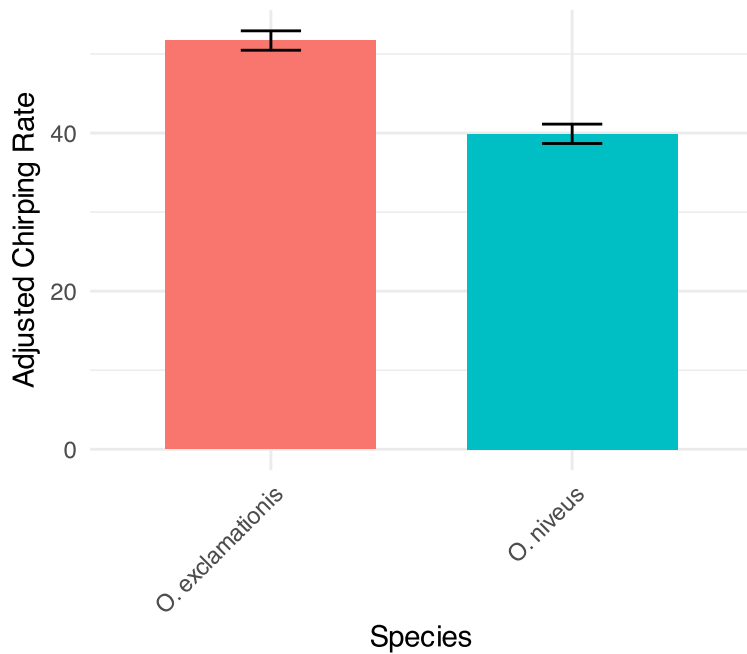
```
# Plot adjusted means with confidence intervals
plot(cricket_adjusted_means, comparisons = TRUE)
```



```
# Bar chart of adjusted means
ggplot(cricket_adj_means_df, aes(x = species, y = emmean, fill = species)) +
  geom_bar(stat = "identity", width = 0.7) +
  geom_errorbar(aes(ymin = lower.CL, ymax = upper.CL), width = 0.2) +
  labs(title = "Adjusted Mean Chirping Rate by Species",
        subtitle = "Means adjusted for temperature",
        x = "Species",
        y = "Adjusted Chirping Rate") +
  theme_minimal() +
  theme(legend.position = "none",
        axis.text.x = element_text(angle = 45, hjust = 1))
```

## Adjusted Mean Chirping Rate by Species

Means adjusted for temperature



## Part 2: Partridge Longevity Analysis

### Data Overview

We'll analyze the effect of mating strategy on male fruitfly longevity, using thorax length as a covariate.

```
# Load the partridge dataset
partridge <- read.csv("data/partridge.csv")

# Create better treatment names
partridge$treatment <- factor(partridge$TREATMEN,
                              levels = 1:5,
                              labels = c("No females",
                                          "One virgin female daily",
                                          "Eight virgin females daily",
                                          "One inseminated female daily",
                                          "Eight inseminated females daily"))

# View data structure
head(partridge)
```

|   | PARTNERS | TYPE | TREATMEN | LONGEV | LLONGEV  | THORAX | RESID1     | PREDICT1 | RESID2      |
|---|----------|------|----------|--------|----------|--------|------------|----------|-------------|
| 1 | 8        | 0    | 1        | 35     | 1.544068 | 0.64   | -5.868456  | 40.86846 | -0.04743024 |
| 2 | 8        | 0    | 1        | 37     | 1.568202 | 0.68   | -9.301196  | 46.30120 | -0.07105067 |
| 3 | 8        | 0    | 1        | 49     | 1.690196 | 0.68   | 2.698804   | 46.30120 | 0.05094369  |
| 4 | 8        | 0    | 1        | 46     | 1.662758 | 0.72   | -5.733936  | 51.73394 | -0.02424867 |
| 5 | 8        | 0    | 1        | 63     | 1.799341 | 0.72   | 11.266064  | 51.73394 | 0.11233405  |
| 6 | 8        | 0    | 1        | 39     | 1.591065 | 0.76   | -18.166676 | 57.16668 | -0.14369601 |

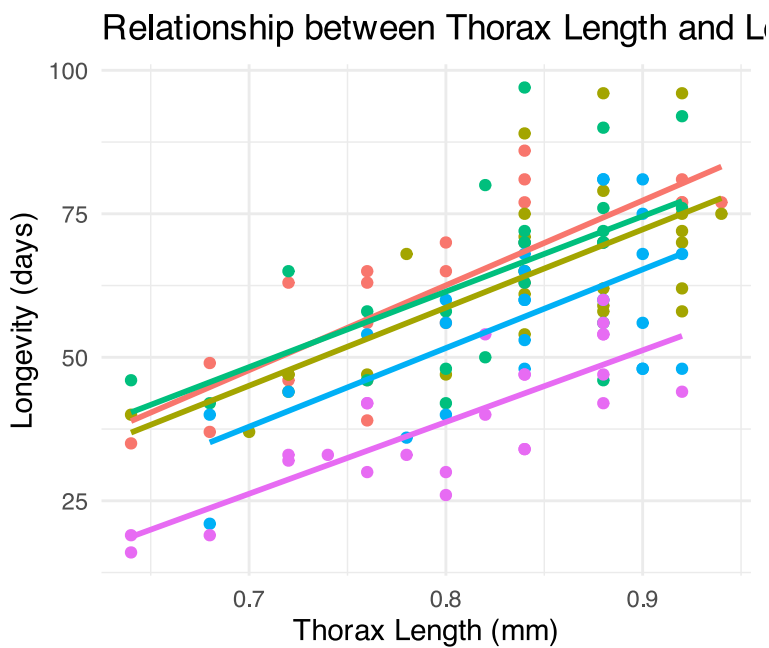
  

|   | PREDICT2 | treatment  |
|---|----------|------------|
| 1 | 1.591498 | No females |
| 2 | 1.639252 | No females |
| 3 | 1.639252 | No females |

```
4 1.687007 No females
5 1.687007 No females
6 1.734761 No females
```

```
# Visualize the relationship between thorax length and longevity by treatment
ggplot(partridge, aes(x = THORAX, y = LONGEV, color = treatment)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Relationship between Thorax Length and Longevity",
       x = "Thorax Length (mm)",
       y = "Longevity (days)",
       color = "Treatment") +
  theme_minimal() +
  theme(legend.position = "bottom")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



virgin female daily    Eight virgin females daily    One inseminated

## Step 1: Test Homogeneity of Slopes

```
# Test for homogeneity of slopes
homo_slopes_model <- lm(LONGEV ~ THORAX * treatment, data = partridge)
Anova(homo_slopes_model, type = 3)
```

Anova Table (Type III tests)

Response: LONGEV

|             | Sum Sq | Df | F value | Pr(>F)           |
|-------------|--------|----|---------|------------------|
| (Intercept) | 755.6  | 1  | 6.6320  | 0.01128 *        |
| THORAX      | 3486.3 | 1  | 30.5999 | 0.0000002017 *** |

```

treatment          36.9   4  0.0810    0.98805
THORAX:treatment   42.5   4  0.0933    0.98441
Residuals          13102.1 115
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

## Step 2: Fit ANCOVA Model

```

# Fit the ANCOVA model (without interaction)
ancova_model <- lm(LONGEV ~ THORAX + treatment, data = partridge)

# Get more detailed summary
summary(ancova_model)

```

```

Call:
lm(formula = LONGEV ~ THORAX + treatment, data = partridge)

Residuals:
    Min       1Q   Median       3Q      Max
-26.189  -6.599  -0.989   6.408  30.244

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    -46.055     10.239  -4.498   ***
THORAX         135.819     12.439  10.919   ***
treatmentOne virgin female daily    -3.929       2.997  -1.311    0.192347
treatmentEight virgin females daily  -1.276       2.983  -0.428    0.669517
treatmentOne inseminated female daily -10.946       2.999  -3.650    0.000391 ***
treatmentEight inseminated females daily -23.879       2.973  -8.031    0.0000000000000783 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.51 on 119 degrees of freedom
Multiple R-squared:  0.6564,    Adjusted R-squared:  0.6419
F-statistic: 45.46 on 5 and 119 DF,  p-value: < 0.0000000000000022

```

```

# View ANOVA table
anova(ancova_model)

```

### Analysis of Variance Table

```

Response: LONGEV
      Df Sum Sq Mean Sq F value    Pr(>F)
THORAX  1 15496.6  15496.6 140.293 < 0.0000000000000022 ***

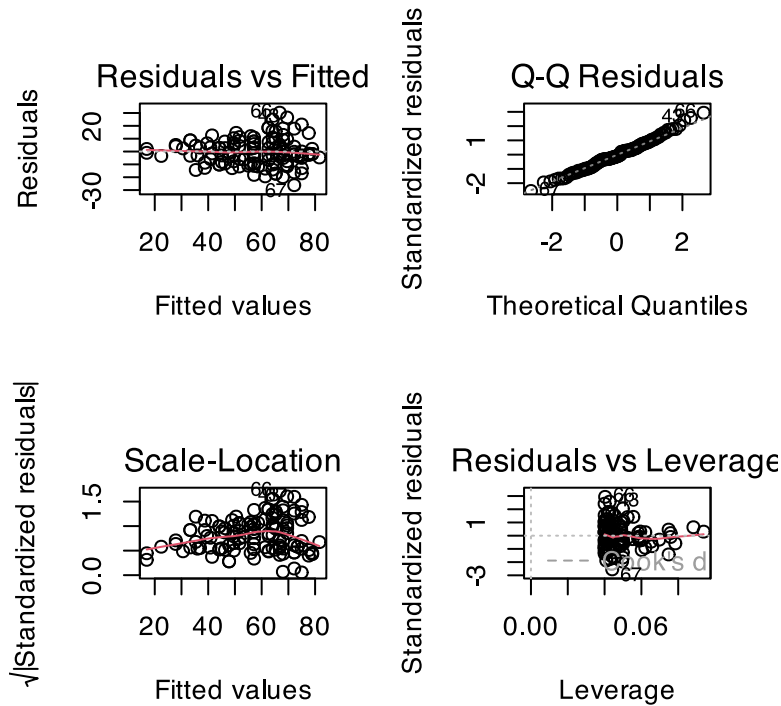
```



```
treatment    4  9611.5  2402.9  21.753    0.0000000000001719 ***
Residuals 119 13144.7    110.5
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

### Step 3: Check Assumptions

```
# Create diagnostic plots
par(mfrow = c(2, 2))
plot(ancova_model)
```



### Step 4: Calculate Adjusted Means

```
# Get adjusted means using emmeans
adjusted_means <- emmeans(ancova_model, "treatment")
adjusted_means
```

| treatment                       | emmean | SE   | df  | lower.CL | upper.CL |
|---------------------------------|--------|------|-----|----------|----------|
| No females                      | 65.4   | 2.11 | 119 | 61.3     | 69.6     |
| One virgin female daily         | 61.5   | 2.11 | 119 | 57.3     | 65.7     |
| Eight virgin females daily      | 64.2   | 2.10 | 119 | 60.0     | 68.3     |
| One inseminated female daily    | 54.5   | 2.11 | 119 | 50.3     | 58.7     |
| Eight inseminated females daily | 41.6   | 2.12 | 119 | 37.4     | 45.8     |

Confidence level used: 0.95

### Step 5: Pairwise Comparisons

```
# Pairwise comparisons of adjusted means
pairs(adjusted_means, adjust = "tukey")
```

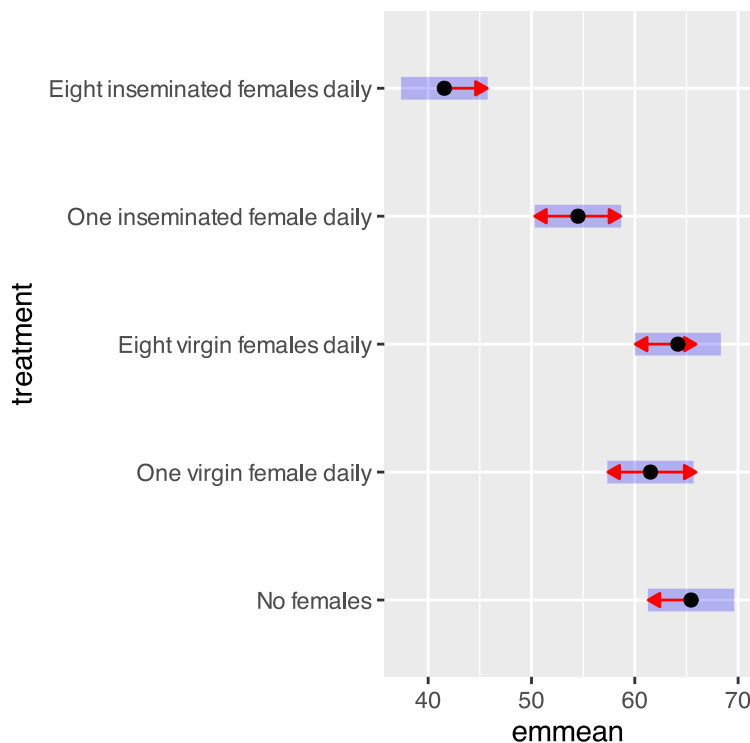
| contrast                                                       | estimate | SE   |
|----------------------------------------------------------------|----------|------|
| No females - One virgin female daily                           | 3.93     | 3.00 |
| No females - Eight virgin females daily                        | 1.28     | 2.98 |
| No females - One inseminated female daily                      | 10.95    | 3.00 |
| No females - Eight inseminated females daily                   | 23.88    | 2.97 |
| One virgin female daily - Eight virgin females daily           | -2.65    | 2.98 |
| One virgin female daily - One inseminated female daily         | 7.02     | 2.97 |
| One virgin female daily - Eight inseminated females daily      | 19.95    | 3.01 |
| Eight virgin females daily - One inseminated female daily      | 9.67     | 2.98 |
| Eight virgin females daily - Eight inseminated females daily   | 22.60    | 2.99 |
| One inseminated female daily - Eight inseminated females daily | 12.93    | 3.01 |

| df  | t.ratio | p.value |
|-----|---------|---------|
| 119 | 1.311   | 0.6849  |
| 119 | 0.428   | 0.9929  |
| 119 | 3.650   | 0.0035  |
| 119 | 8.031   | <.0001  |
| 119 | -0.891  | 0.8996  |
| 119 | 2.361   | 0.1336  |
| 119 | 6.636   | <.0001  |
| 119 | 3.249   | 0.0129  |
| 119 | 7.560   | <.0001  |
| 119 | 4.298   | 0.0003  |

P value adjustment: tukey method for comparing a family of 5 estimates

```
# Plot adjusted means with confidence intervals
plot(adjusted_means, comparisons = TRUE)
```



### Part 3: Example with Heterogeneous Slopes

Let's look at an example where slopes are NOT homogeneous using sea urchin data.

```

# Create simulated sea urchin data with heterogeneous slopes
set.seed(345)
n <- 72 # 24 urchins per group

# Create data frame
treatments <- rep(c("Initial", "Low Food", "High Food"), each = n/3)
volume <- c(
  runif(n/3, 10, 40), # Initial
  runif(n/3, 10, 40), # Low Food
  runif(n/3, 10, 40)  # High Food
)

# Create suture width with different slopes for each treatment
suture_width <- ifelse(
  treatments == "Initial", 0.05 + 0.002 * volume,
  ifelse(
    treatments == "Low Food", 0.04 + 0.0005 * volume,
    0.02 + 0.003 * volume # High Food
  )
) + rnorm(n, 0, 0.01)

urchin_data <- data.frame(treatment = treatments, volume = volume, suture_width = suture_width)

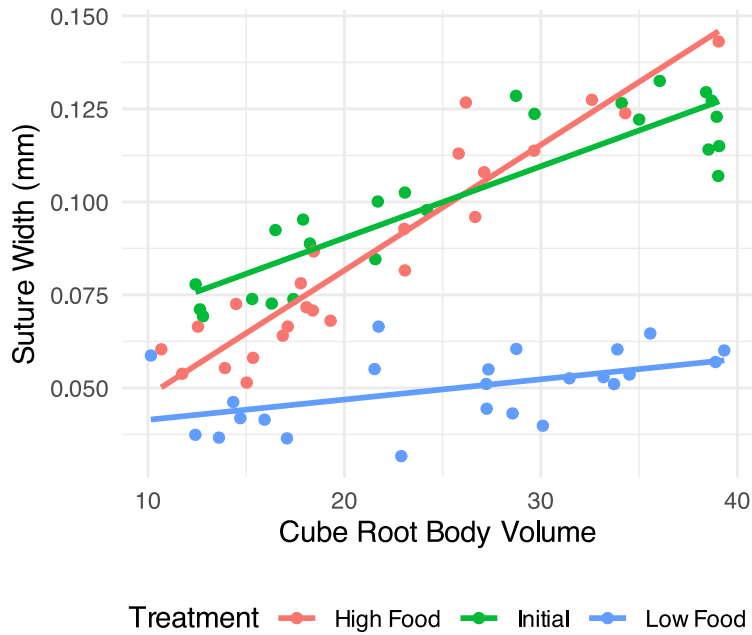
# Plot the data with regression lines
ggplot(urchin_data, aes(x = volume, y = suture_width, color = treatment)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Sea Urchin Suture Width vs. Volume",
       subtitle = "Example with Heterogeneous Slopes",
       x = "Cube Root Body Volume",
       y = "Suture Width (mm)",
       color = "Treatment") +
  theme_minimal() +
  theme(legend.position = "bottom")

```

```
`geom_smooth()` using formula = 'y ~ x'
```

## Sea Urchin Suture Width vs. Volume

Example with Heterogeneous Slopes



## Test for Homogeneity of Slopes

```
# Fit model with interaction
urchin_model <- lm(suture_width ~ volume * treatment, data = urchin_data)
Anova(urchin_model, type = 3)
```

Anova Table (Type III tests)

```
Response: suture_width
```

|                  | Sum Sq    | Df | F value | Pr(>F)                    |
|------------------|-----------|----|---------|---------------------------|
| (Intercept)      | 0.0005253 | 1  | 5.91    | 0.01778 *                 |
| volume           | 0.0151663 | 1  | 170.64  | < 0.00000000000000022 *** |
| treatment        | 0.0020070 | 2  | 11.29   | 0.00006064438398 ***      |
| volume:treatment | 0.0062129 | 2  | 34.95   | 0.000000000004453 ***     |
| Residuals        | 0.0058662 | 66 |         |                           |

---  
 Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**Result:** With  $p < 0.05$ , we have heterogeneous slopes! Standard ANCOVA would be inappropriate here.

## What to do with Heterogeneous Slopes

When slopes are not homogeneous, you have several options:

```
# Option: Analyze groups separately
initial_model <- lm(suture_width ~ volume, data = filter(urchin_data, treatment == "Initial"))
low_food_model <- lm(suture_width ~ volume, data = filter(urchin_data, treatment == "Low Food"))
high_food_model <- lm(suture_width ~ volume, data = filter(urchin_data, treatment == "High Food"))

# Summary for each group
initial_model
```

```
Call:
lm(formula = suture_width ~ volume, data = filter(urchin_data,
  treatment == "Initial"))
```

```
Coefficients:
(Intercept)      volume
  0.051785      0.001926
```

low\_food\_model

```
Call:
lm(formula = suture_width ~ volume, data = filter(urchin_data,
  treatment == "Low Food"))
```

```
Coefficients:
(Intercept)      volume
  0.0359532      0.0005453
```

high\_food\_model

```
Call:
lm(formula = suture_width ~ volume, data = filter(urchin_data,
  treatment == "High Food"))
```

```
Coefficients:
(Intercept)      volume
  0.014077      0.003376
```

## Summary Checklist for ANCOVA

When conducting ANCOVA, always follow these steps:

### 💡 ANCOVA Checklist

1. **Visualize your data** - plot response vs covariate, colored by groups
2. **Test homogeneity of slopes** - fit model with interaction term
  - If  $p > 0.05$ : proceed with ANCOVA
  - If  $p < 0.05$ : use alternative approaches
3. **Fit ANCOVA model** - response ~ covariate + factor
4. **Check assumptions** - use diagnostic plots
5. **Interpret results** - focus on adjusted means, not raw means
6. **Conduct post-hoc tests** - pairwise comparisons if needed
7. **Visualize results** - show adjusted means with confidence intervals

## Key Points to Remember

- **ANCOVA increases power** by accounting for covariate variation
- **Adjusted means** are what we compare, not raw group means

- **Homogeneity of slopes** is the most critical assumption
- **Parallel lines** in your plot suggest homogeneous slopes
- **Non-parallel lines** indicate heterogeneous slopes - use alternative methods

#### ! Key Points from ANCOVA Analysis

1. **Test homogeneity of slopes first** - this is the most critical assumption
2. **ANCOVA compares adjusted means** at the mean value of the covariate
3. **Increases statistical power** by removing variation due to the covariate
4. **Choose appropriate methods** based on whether slopes are homogeneous
5. **Visualize your results** clearly showing the relationship between variables
6. **Check all assumptions** using diagnostic plots
7. **Interpret in biological context** - what do the adjusted means tell us?

Remember: The covariate should be measured independently of the treatment and should not be affected by the treatment itself!